

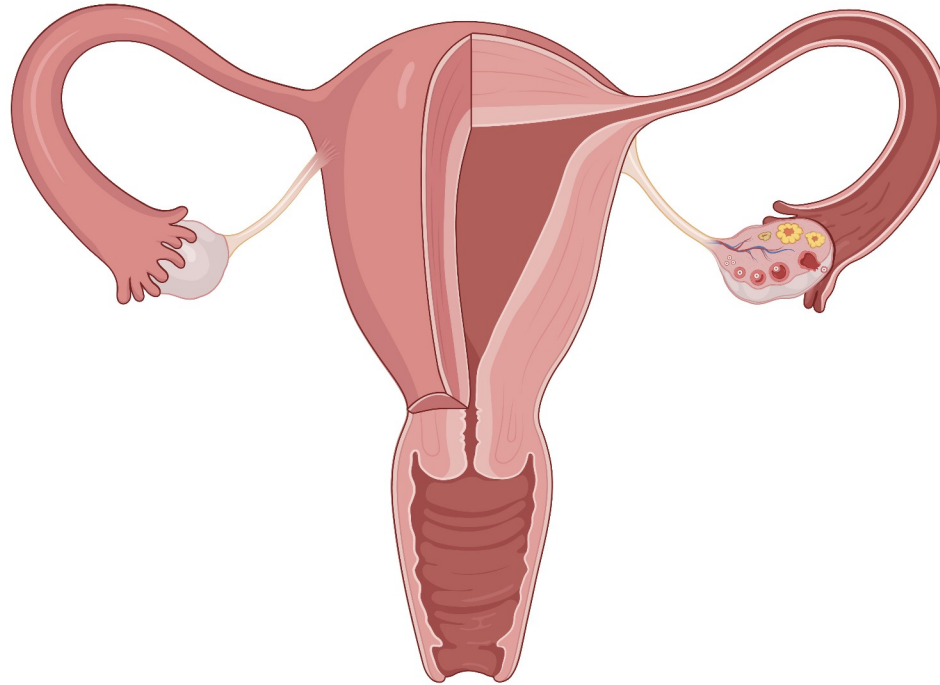
What is AUB?

Causes

Work-up

Cases

Abnormal Uterine Bleeding



BONUS: HPO axis

What is AUB?

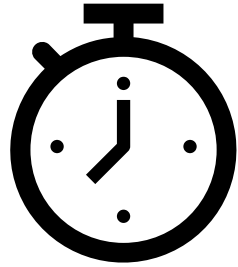
Causes

Work-up

Cases

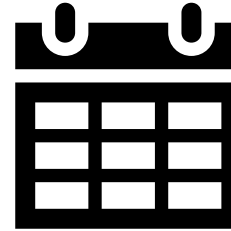
What is abnormal uterine bleeding (AUB)?

Duration



Menses lasting >8 days

Regularity



High variation in cycle length.

Generally, cycle length should vary by:

- ≤ 9 d if age <25 or > 45
- ≤ 7 d if age 26 - 41

Frequency



Absent
(**amenorrhea**)
No menses x 3 mos

Infrequent
(**oligo-**)
Cycle > 35 days

Frequent
(**poly-**)
Cycle < 21 days

Volume



Subjective: Interferes with patient's life, job, well-being, etc.

Objective:

- Changing pad/tampon q2h
- Changing pad/tampon overnight
- Symptomatic

What is AUB?

Causes

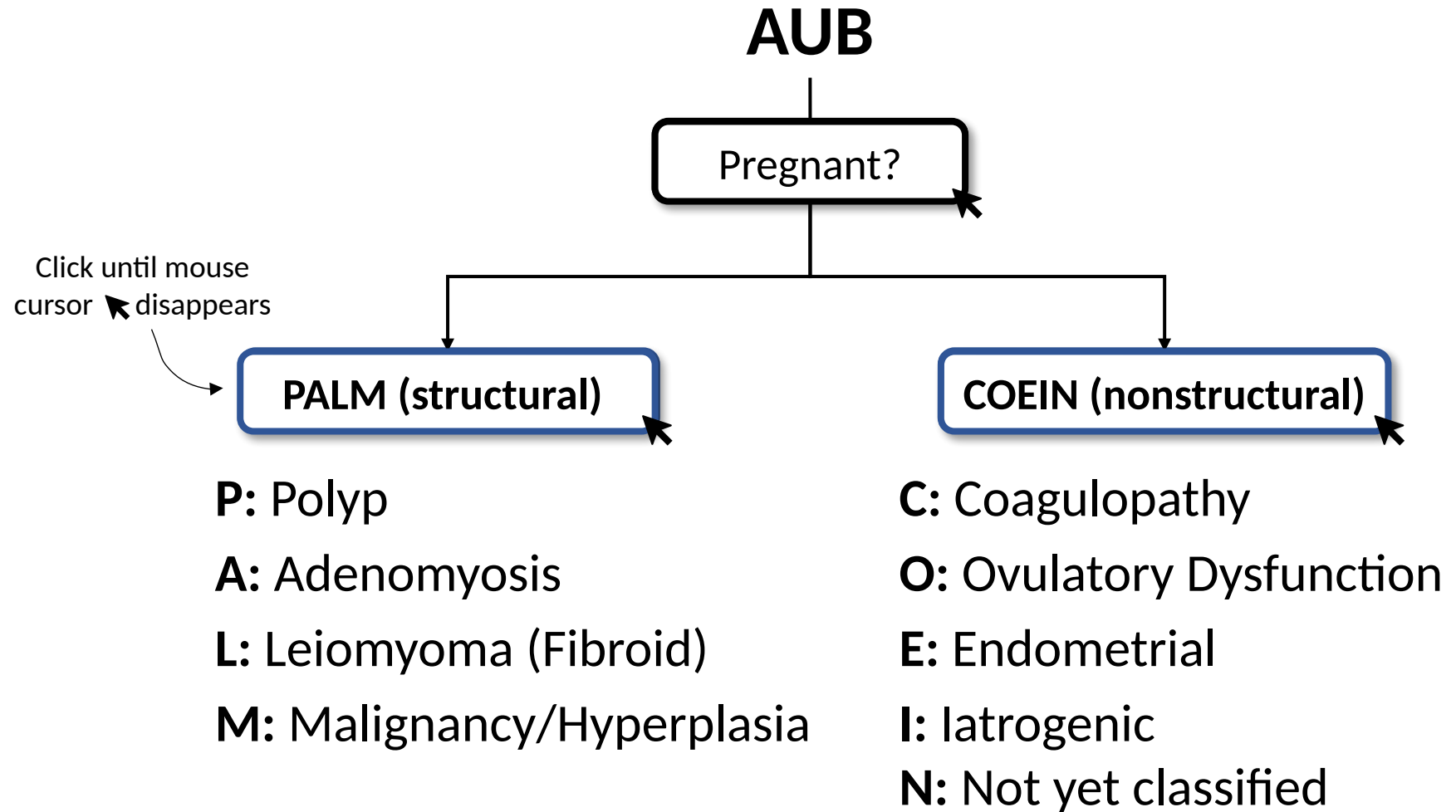
Work-up

Cases

BONUS: HPO axis

What causes AUB?

Use the PALM-COEIN mnemonic to develop a differential



What is AUB?

Causes

Work-up

Basic Workup

Structural

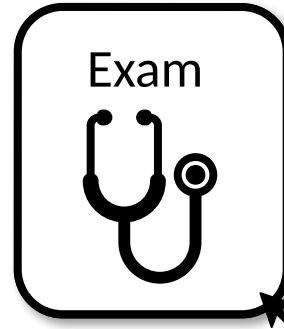
Functional

Cases

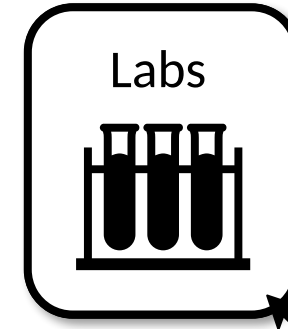
BONUS: HPO axis

How do you work-up AUB?

All patients with AUB
require the following
basic tests:



Pelvic Exam



Pregnancy Test
CBC
Iron studies

What is AUB?

Causes

Work-up

Basic Workup

Structural

Nonstructural

Cases

How do you work-up AUB?

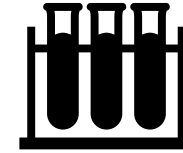
All patients with AUB
require the following
basic tests:

Exam



Pelvic Exam

Labs



Pregnancy Test
CBC
Iron studies

Further work-up of **structural** causes?



Transvaginal ultrasound



± Endometrial biopsy

Further work-up of **nonstructural**
causes?



Additional History

Who should receive an
endometrial biopsy?

Risk factors for hyperplasia

- peri- or post-menopause
- chronic anovulation
- unopposed estrogen exposure
- tobacco use

How do we initiate cause-
specific workups?

What is AUB?

Causes

Work-up

Basic Workup

Structural

Nonstructural

Cases

BONUS: HPO axis

How do you work-up functional causes of AUB?

Patient Presentation

Very heavy periods since menarche, soaking through pads. Mom has recurrent nosebleeds.

Irregular menses since menarche with insulin resistance and facial hair.

Pain with intercourse + irregular, light bleeding and vaginal discharge

Recently placed Nexplanon with irregular spotting.

Clues

Clues to diagnosis?

Clues to diagnosis?

Clues to diagnosis?

Clues to diagnosis?

Evaluation

Next steps in evaluation?

Next steps in evaluation?

Next steps in evaluation?

Next steps in evaluation?

What is AUB?

Causes

Work-up

Basic Workup

Structural

Nonstructural

Cases

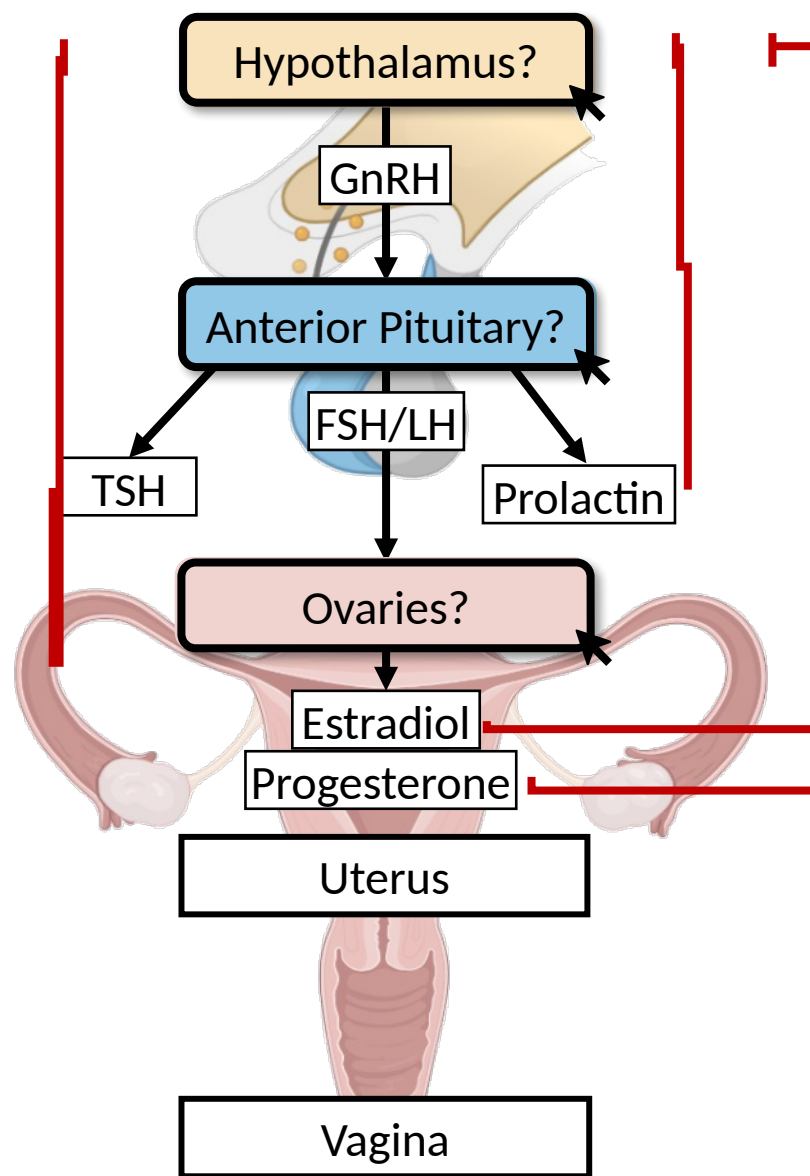
Hypothalamus secretes GnRH which acts on the anterior pituitary (AP)

(AP) secretes TSH, FSH/LH, and prolactin.

FSH/LH acts on the ovaries to secrete estradiol/progesterone.

BONUS: HPO axis

Hypothalamic-Pituitary-Ovarian Axis



How does hypothyroidism cause menstrual irregularities?

How does the physiology of menopause explain the elevated FSH we see in perimenopausal patients?

What is AUB?

Causes

Work-up

Basic Workup

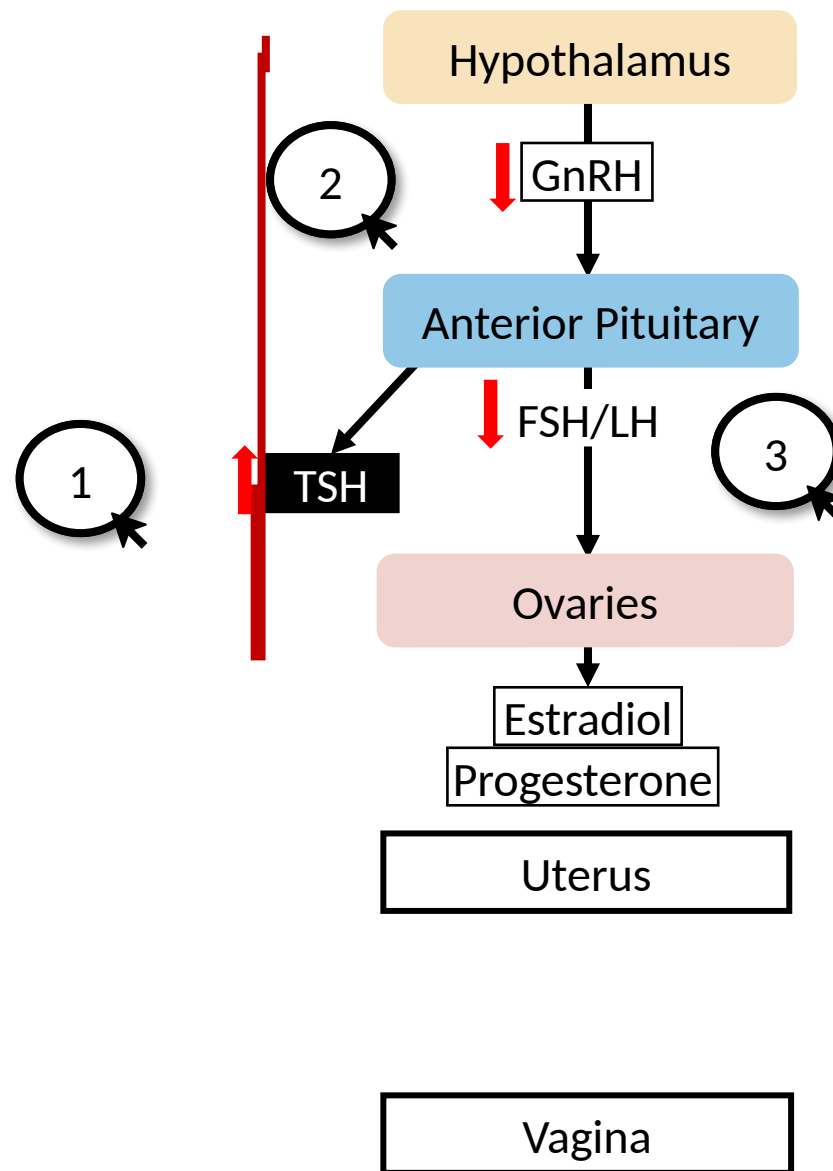
Structural

Functional

Cases

BONUS: HPO axis

Hypothalamic-Pituitary-Ovarian Axis



How does hypothyroidism cause menstrual irregularities?

Hypothyroidism → ↑ TSH levels

TSH provides negative feedback to hypothalamus → ↓ GnRH

↓ GnRH → ↓ FSH/LH ovarian → ↓ stimulation to ovulate

How does the physiology of menopause explain the elevated FSH we see in perimenopausal patients?

What is AUB?

Causes

Work-up

Basic Workup

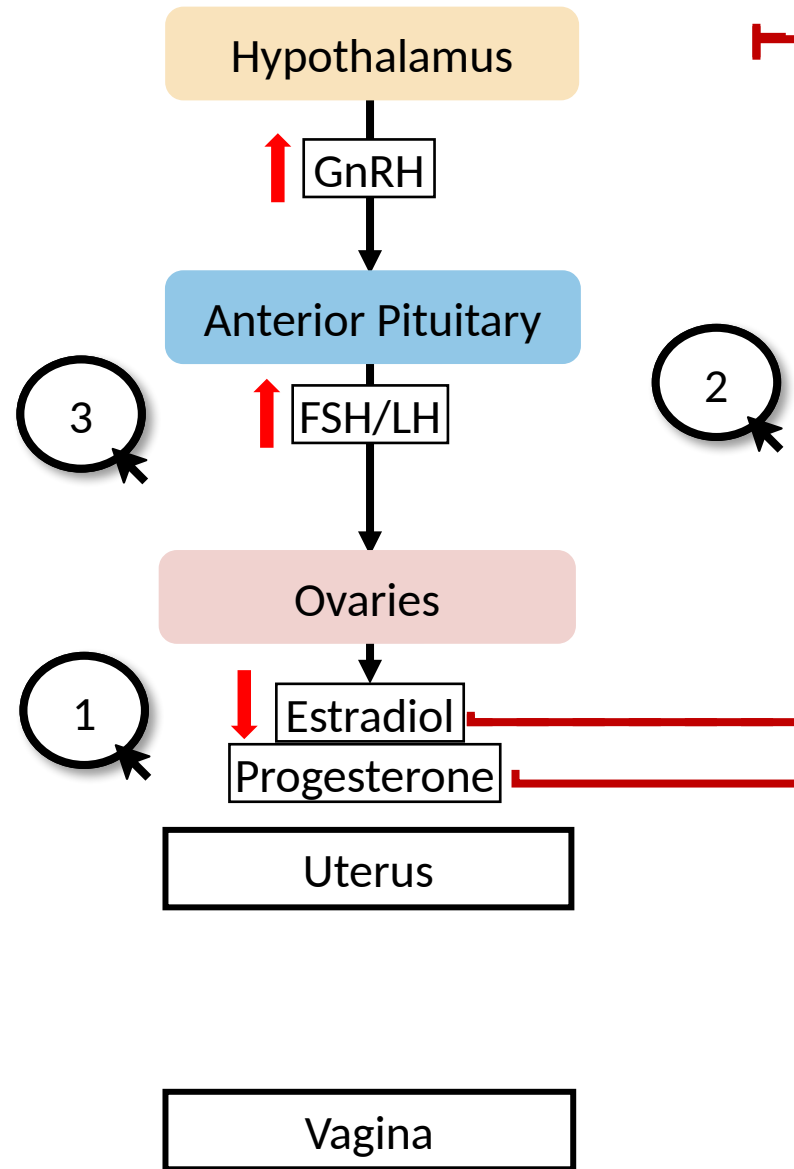
Structural

Functional

Cases

BONUS: HPO axis

Hypothalamic-Pituitary-Ovarian Axis



How does hypothyroidism cause menstrual irregularities?

How does the physiology of menopause explain the elevated FSH we see in perimenopausal patients?

Peri-menopause is due to ↓ estrogen production at the ovary

↓ estrogen → ↑ GnRH production at the hypothalamus

↑ GnRH → ↑ FSH/LH

What is AUB?

Causes

Work-up

Cases

Case 1

32 yo F presenting with self-reported "heavy periods" for the past 12 mos. She has a period every 26-28 days that lasts 8 -9 days. She soaks through a pad every 2 hours on heavy days and has had blood soak through her pants. Her periods previously lasted 4-5 days and were not as heavy. No other major changes in the last year.

Pelvic exam: mildly enlarged uterus and no other major abnormalities.

Labs: iron deficiency without anemia, negative pregnancy test.

Is this abnormal uterine bleeding (AUB)?

Yes. It is heavy (high volume) and prolonged (long duration).

What is the most likely cause of AUB?

Structural cause, most likely fibroids or adenomyosis considering physical exam findings.

What additional evaluation should be considered?

Transvaginal ultrasound

Case 2

39 yo F presents for "irregular periods". She reports that for the last 2 years, she has been having increasingly infrequent periods occurring every 40-60 days. Her menstrual bleeding is variable in volume. She has a family history of celiac disease and lupus.

ROS: fatigue, hair loss, and an unintentional 15-pound weight gain over the last 2 years.

Pelvic exam is unremarkable.

Labs: negative pregnancy test, normal blood counts, coagulation markers, no iron deficiency.

What is her bleeding pattern?

Infrequent (oligomenorrhea, cycles > 35d) and irregular (variable cycle length)

What is the most likely cause of AUB?

Ovulatory dysfunction, most likely from endocrine causes (e.g., hypothyroidism - fatigue, hair loss, weight gain). Given age, premature ovarian failure can be considered.

What is your next step in evaluation?

Pregnancy test, TSH, prolactin, ± FSH. Consider total testosterone if risk factors for PCOS.

2018 FIGO Update & Currency Note (verified June 2026)

Standing framework is current: the FIGO PALM-COEIN system (Munro, Int J Gynecol Obstet 2018 revision) and the AUB definitions taught here remain the accepted standard. No change to the teaching.

Correction made to this deck: the "I" in PALM-COEIN is IATROGENIC, not "Idiopathic" — i.e., hormonal contraception (IUDs, implants such as Nexplanon), anticoagulants, SERMs (tamoxifen/raloxifene), and dopamine antagonists. ("Idiopathic" was a transcription slip; FIGO and the lesson script both say Iatrogenic.)

Framing reminders worth adding at the board:

FIGO normal-cycle limits: frequency 24–38 d, duration ≤ 8 d, cycle-to-cycle variation ≤ 7 –9 d (age-dependent).

Any post-menopausal bleeding is abnormal until proven otherwise → transvaginal US $\pm$ endometrial biopsy to exclude endometrial cancer.

Levonorgestrel IUD and tranexamic acid are first-line medical options for heavy menstrual bleeding from non-structural / fibroid causes (ACOG).

Take-Home Points

1. AUB is defined by abnormality in any of four parameters: duration (>8 d), frequency (<24 or >38 d), regularity (variation $>7-9$ d by age), or volume (interferes with life / objective soaking).
2. After ruling out pregnancy, use PALM-COEIN — PALM = structural (Polyp, Adenomyosis, Leiomyoma, Malignancy/hyperplasia), COEIN = non-structural (Coagulopathy, Ovulatory dysfunction, Endometrial, Iatrogenic, Not yet classified).
3. Every patient with AUB gets a pelvic exam plus three labs: pregnancy test, CBC, and iron studies (ferritin <50 ng/mL is low in pre-menopausal patients).
4. Add transvaginal ultrasound for structural work-up; endometrial biopsy for hyperplasia/cancer risk (age >45 , post-menopausal bleeding, chronic anovulation/obesity, unopposed estrogen, tamoxifen).
5. ~20% of heavy menstrual bleeding (higher in adolescents) reflects a bleeding disorder — screen with vWF panel/PT/PTT when bleeding dates to menarche or there is a personal/family bleeding history.

Attribution, License & Optimization Notes

Original: "Abnormal Uterine Bleeding," TeachIM.org. Authors: Nikki Zarling MD, Natalie deQuillfeldt MD, Brandon Fainstad, Molly Brett MD; sec. ed. Adelaide McClintock MD; exec. ed. Yilin Zhang MD. Published April 2022.

teachim.org/teaching_material/abnormal-uterine-bleeding/ — CC BY-NC 4.0 (attribution, non-commercial). Educational use only; clinician-reviewed before teaching.

Optimization (June 2026):

Correction (surgical, all click-to-reveal animations & graphics preserved): PALM-COEIN "I: Idiopathic" → "I: Iatrogenic" on slide 3 (mnemonic legend) and slide 6 (the Nexplanon vignette reveal label). Matches FIGO 2018 and the lesson's own teaching script.

Added: this 2018-FIGO currency slide, take-home slide; HTML package (pre-work, ASK points, click-to-reveal cases), 5 MCQs, 17 Anki cards, EMR dotphrase, one-pager PDF.

Companions copied & linked (reviewed, unchanged): Teaching Guide (AUB-Teach-IM-Guide-final.docx) + Learner Worksheet (Learner-Worksheet-1.pdf).

All 11 original slides (1 hidden worksheet) preserved → 14 total.